

DWIJ JOSHI

Mechatronics Engineering Student · Embedded Systems · Software Testing & QA
Scarborough, ON · 437-925-2502 · dwijjoshi14@gmail.com · linkedin.com/in/dwij-joshi-14-tron · github.com/dwijjoshi14 · dwijjoshi14.github.io

PROFESSIONAL SUMMARY

Mechatronics Engineering student at Toronto Metropolitan University with hands-on experience building embedded control systems, writing and validating C++ firmware, and documenting technical outcomes. Background spans hardware-software integration, software QA, and data-driven program support. Comfortable owning work end-to-end design, test, debug, document. Seeking a technical co-op or internship where precision engineering and real-world impact are the standard.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL

Tools & Platforms: Git, Arduino IDE, PlatformIO, MATLAB, Tinkercad, VS Code, Microsoft Excel/Word/PowerPoint

Testing & QA: Test case design, black-box & white-box testing, defect tracking, validation & verification

Embedded Systems: PWM control, ADC sensor integration, PMOS driver circuits, real-time feedback loops, UART/I2C/SPI

Methodologies: Agile, Waterfall, SDLC, technical documentation

ENGINEERING PROJECTS

RTOS Task Scheduler — FreeRTOS Temperature Control System | C11 · FreeRTOS API · POSIX pthreads · Queue · Semaphore · Mutex ·

- Implemented a three-task FreeRTOS system (SensorTask, ControlTask, DisplayTask) with priority-based scheduling; ControlTask assigned highest priority to guarantee sub-cycle response to temperature threshold crossings.
- Designed inter-task communication using a typed queue (SensorTask → ControlTask), a binary semaphore for critical alert signalling (ControlTask → DisplayTask), and a mutex to protect shared console output across all three tasks.
- Built a POSIX simulation layer mapping the full FreeRTOS API (xTaskCreate, xQueueSend, xSemaphoreGive/Take) to pthreads, code is API-identical to STM32 deployment; porting requires changing three HAL calls only.

Embedded Heat Management & Control System | C++ · Arduino · PWM · TMP36 · NeoPixel · Tinkercad

- Built a closed-loop thermal control system on Arduino Uno: TMP36 ADC sensing, threshold-based PWM fan control via PMOS high-side driver, simulated in Tinkercad Circuits.
- Identified and resolved inverted PWM logic error caused by PMOS high-side configuration; fix required inverting analogWrite output (255 → fanSpeed) to correct fan behavior.
- Delivered synchronized multi-output feedback: 16x2 LCD (live temp + fan %), 80-LED NeoPixel strip (color-coded thermal zones), and PWM-controlled cooling fan.
- Validated threshold decision logic in a standalone C++ file independently of the Arduino hardware environment, enabling unit-level algorithm testing before hardware integration.

DC Circuit Analyzer | C++17 · OOP · KVL/KCL · Ohm's Law · Make

- Built a command-line tool that applies Ohm's Law and Kirchhoff's Laws to analyze series and parallel DC resistive circuits, computing total resistance, branch currents, voltage drops, and power.
- Implemented automatic KVL/KCL self-verification after each analysis run, catching computational errors without manual calculation.
- Structured using a Circuit class and Resistor data type with full input validation and exception handling.

WORK EXPERIENCE

Administrative Assistant | Rosalie Hall Mental Health Institution — Toronto, ON

Jul 2024 – Aug 2025

- Rebuilt Excel-based client tracking for 50+ users, cutting report generation from ~3 hours to under 45 minutes and reducing manual entry errors by 35%.
- Designed a standardized data-entry protocol adopted across 3 program teams, improving cross-department consistency and audit readiness.
- Produced monthly outcome dashboards for clinical leadership, translating raw data into actionable summaries informing staffing and resource decisions.

Program Assistant | Leacock Foundation — Toronto, ON

Feb 2023 – Oct 2025

- Tracked engagement for 100+ students across 4 streams; identified a 15% attendance drop-off trend prompting a schedule change that recovered participation within 6 weeks.
- Redesigned evaluation documentation workflow, reducing end-of-cycle report compilation by 30%.

League Ambassador | JAM Sports — Toronto, ON

Sept 2022 – Dec 2022

- Coordinated logistics and real-time issue resolution for 30+ participants per session; zero-incident operational record.

EDUCATION

Bachelor of Engineering — Mechatronics Engineering · Toronto Metropolitan University · Expected: April 2029

Relevant Coursework: Data Structures & Algorithms, Embedded Systems, Digital Logic, Circuit Analysis, OOP, Discrete Mathematics, Engineering Communication